Predicting downstream capabilities

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

The blog's §9 explains why predicting cross-entropy is easier than predicting task accuracy. Accuracy depends on threshold decisions and competing answers. If every small model scores near zero, fitting a smooth accuracy curve can produce an impressive-looking but unsupported forecast.

Three routes

Route	Intermediate quantity	Main condition
General loss to accuracy	Common validation loss	Stable relation to the task
Task NLL to accuracy	Correct-answer likelihood	Calibrated likelihood-to-score relation
Difficulty-cluster prediction	Per-item pass-rate vectors	Stable clusters and informative difficulty range

For a task answer's NLL z, a possible calibration is $p(z) = g + (1 - g)/(1 + \exp[a(z - b)])$, with guessing floor g, positive slope parameter a, and midpoint b. This is an illustrative calibration family, not a claim that every task follows it.

Procedure

1. Freeze benchmark version, templates, scoring masks, and sampling budget.
2. Measure continuous answer NLL or BPB alongside accuracy on each ladder checkpoint.
3. Fit size/tokens-to-task-loss on fitting tiers, then calibrate task-loss-to-accuracy on development data.
4. For floor-limited tasks, consider COD-style difficulty clustering using repeated samples across small models. Freeze clusters before final target evaluation.
5. Test the entire prediction pipeline on larger held-out models. Propagate uncertainty through both stages.
6. Reject the accuracy forecast when residuals exceed the decision tolerance; retain the continuous metric if it remains useful.

The OLMo Task Ladder is a concrete reference for two-stage prediction. COD is a different approach built around difficulty subsets and a mapping back to the full benchmark. Neither removes the need to test architecture/data transfer or to account for benchmark sampling variance.

Deliverable: per-task forecast with calibration data, uncertainty, holdout error, and a flag for tasks with insufficient signal.

Sources: blog §9; Bhagia et al., 2024, OLMo Task Ladder; COD, 2026 version; Schaeffer et al., 2024.

Blog/notes crosswalk · Detailed companion